

Body Mass Index and Kidney Disease-Related Mortality in Midlife: A Nationwide Cohort of 2.3 Million Adolescents

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Objective: This study aimed to evaluate the association of body mass index (BMI) in adolescence with mortality attributed to kidney disease.

Methods: In this study, 2,294,139 Jewish Israeli adolescents with measured weight and height at 17 years old during the military fitness assessment were analyzed with a follow-up extending up to 45 years. All kidney-related outcomes, coded by the Central Bureau of Statistics from death notifications as the underlying cause of death, were obtained by linkage. Cox hazards models were applied.

Results: During 42,297,007 person-years of follow-up (median 18.4 years), 226 deaths related to kidney disease were recorded. There was an increased risk for kidney-related death among adolescents with overweight and obesity with adjusted hazard ratios of 2.7 (95% CI: 1.8-3.9) and 8.4 (5.1-13.8), respectively, with BMI between 18.5 and 22.0 kg/m² as the reference. A 15% increased risk for kidney-related mortality (1.11-1.19) per unit increment in BMI was observed. Furthermore, a multivariable spline model indicated a minimum risk for kidney-related mortality starting at BMI of 18.6 kg/m² with significantly increased risk seen above values of 22.8 kg/m². The results withstood extensive sensitivity analyses, including stratification of kidney-related death attributed to acute, chronic, and total kidney disease.

Conclusions: Adolescent overweight and obesity are risk markers for kidney-related mortality over 4 decades.

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Introduction

Obesity is increasingly evident in childhood and adolescence and is recognized as a global epidemic in the Western world. Its prevalence has tripled in many countries worldwide with a reported prevalence of more than 30% in the United States and other developed countries (1). Recent evidence from 2015 has suggested that chronic kidney disease (CKD) is the second leading cause of body mass index (BMI)-related disability-adjusted life-years (2). In addition, according to Global Burden of Diseases data, mortality attributed to CKD among young adults (<50 years) has increased continuously in parallel to adolescent obesity (3). In the United States, for instance, an annual increase of 1.6% was recorded between 1995 and 2015 for young adults. On the contrary, several reports have suggested that obesity does not confer a risk for future CKD (4) and may even play a protective role in certain settings (a phenomenon also referred to as the “obesity paradox”) (5).

In either case, most investigations on the association between obesity and CKD have been limited to adulthood obesity (6–8). There are

limited data regarding the association between adolescent or childhood BMI and future mortality risk related to kidney disease, as opposed to the well-established association with coronary heart disease outcomes (9,10). The few investigations that have included adolescents and documented an association between adolescent obesity and future CKD risk used the World Health Organization (11) or the U.S. Centers for Disease Control and Prevention (12) classifications. These classifications include a broad normal BMI range as a reference and, consequently, may attenuate the risk attributed to obesity (13) or mask the BMI cutoff that is associated with increased risk. Notably, these estimates may well be biased by the use of self-reported and recalled data (11).

We previously reported on the association between measured BMI at adolescence and the future risk of cardiovascular- (14) and diabetes-related (13) death in midlife. For both death causes, BMI above 20 kg/m² at age 17 was associated with increased mortality risk. In the current report, we elucidate the association between measured adolescent BMI

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and death attributed to kidney disease in a nationwide cohort of 2.3 million adolescents.

Methods

Study population

The study sample comprised all 2,454,693 Israeli teenagers who underwent medical assessments at ages 16 to 19 years between January 1, 1967, and December 31, 2010, prior to military conscription, regardless of future drafting into service. Participants with missing weight and height data ($n = 64,186$) and those belonging to non-Jewish minorities ($n = 92,377$) were excluded from analysis, as they are considered unrepresentative of their source population (14). We also excluded 3,991 participants who died before 1981, the year from which the underlying cause of death was available to us (Figure 1). Included in the final study sample were 2,294,139 participants, of whom 40.4% were women. Ethical approval for the study was provided by the Israel Defense Forces Medical Corps Institutional Review Board, which waived the need for informed consent based on the strict maintenance of participants' anonymity.

Primary outcomes and documentation of the cause of death

The primary outcome of the study was kidney-related mortality coded as the underlying cause (*International Classification of Diseases, Ninth Revision* [ICD-9], 580-589; *International Classification of Diseases, Tenth Revision* [ICD-10], N00-N29). Renal cancer-related deaths were not included as an outcome. The underlying cause of

death, as routinely officially coded from death notifications by the Israel Central Bureau of Statistics according to the ICD-9 and ICD-10, was linked by using the participants' national identification number. The cause of death was available to us only from 1981 and onward. The occurrence of death among recruits on active duty or in reserve service is documented in Israel Defense Forces records since 1967 with an indicator of deaths caused by service-related trauma (no other causes are provided). Follow-up ceased on the date of death from any cause or June 30, 2011, whichever came first.

Data collection and study variables

The health examination was performed by military physicians who also reviewed the previous medical history provided by the family physician. Standardized measurements of weight and height were undertaken by trained personnel with each participant barefoot and in underwear; BMI was calculated. Data regarding socioeconomic status, education, country of origin, and country of birth were recorded. Education was dichotomized into those with 12 versus ≤ 11 years of formal schooling (15). Socioeconomic status, based on the locality of residence at the time of examination (14), was grouped into low, medium, and high. Country of origin (classified as the birthplace of the father or grandfather, if the father was born in Israel) and country of birth were grouped as reported elsewhere (14,15). Blood pressure measurements at rest in the sitting position were collected from 1967 but were computerized only from 1977 onward. These measurements were conducted with an appropriately sized cuff on the right arm using a manual sphygmomanometer as detailed elsewhere (16).

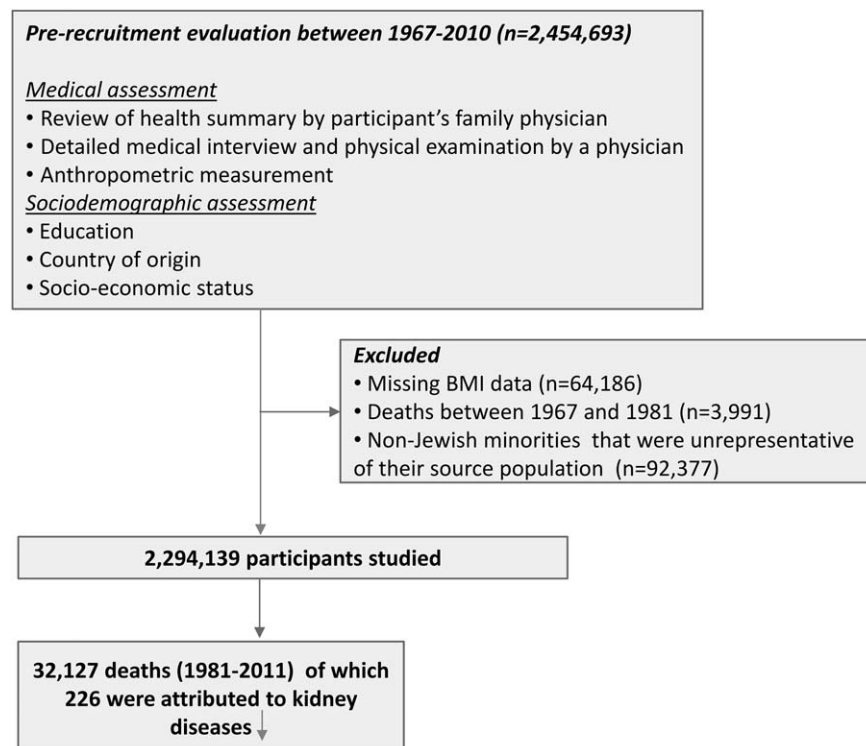


Figure 1 Schematic diagram of the study design. Note that nearly 80% ($n = 3,188$) of the 3,991 deaths that occurred before 1981 were attributed to military trauma.

Statistical analysis

Adolescents in our study were expected to have completed more than 98.5% of their growth (17), thereby permitting the use of absolute BMI measures as a continuous or categorical variable. We grouped BMI categories according to a modified classification of the World Health Organization that included the division of the normal range to low-normal ($18.50 \leq \text{BMI} < 21.99$) and high-normal ($22.0 \leq \text{BMI} < 24.99$). The abnormal BMI subgroups included underweight ($\text{BMI} < 18.50$), overweight ($25.0 \leq \text{BMI} < 29.99$), and obesity ($\text{BMI} \geq 30$). Person-year mortality rates were calculated with follow-up commencing from 1981. Cox proportional hazard models were used to estimate the hazard ratios (HRs) and 95% CIs for kidney mortality with the low-normal group as the reference category. The following two main prespecified models were applied: unadjusted and a multivariable model adjusted for age, sex, year of birth, education, and country of origin. Variables in multivariable models were included only if they were associated with kidney mortality. In the absence of interaction between sex and BMI for kidney mortality ($P > 0.6$ for all models), we analyzed men and women grouped together.

Several sensitivity analyses were conducted. We restricted the Cox analysis to those with an unimpaired medical status at baseline (i.e., no indication of any medical diagnosis in the prerecruitment medical evaluation that would limit eligibility for combat service) as reported previously (14,18) to avert confounding by a preexisting medical diagnosis, such as diabetes, hypertension, history of kidney injury, or rheumatic disease. The analysis was also stratified by deaths occurring prior to or later than age 50 (median age of death was 49.6 years), by deaths occurring before or later than 2006 (the

median year of CKD deaths), and by deaths attributed to acute (ICD-9: 584, ICD-10: N17; $n = 59$ deaths), chronic (ICD-9: 585, ICD-10: N18; $n = 117$), and unspecified reason of kidney injury (ICD-9: 586, ICD-10: N179; $n = 44$). We conducted a sensitivity analysis limited to those examined from 1977 onward (the period for which computerized blood pressure data were available) and included mean blood pressure ($[2 \times P_{\text{Diastolic}} + P_{\text{Systolic}}]/3$) as a continuous variable in the multivariable model. Multiple imputation was applied to those with missing data (1.4% of examinees) as described previously (14). Analyses were performed by using SPSS Statistics (version 23.0; IBM Corp., Armonk, New York).

Results

Table 1 summarizes the characteristics of the study participants. The mean age at enrollment was 17.3 ± 0.4 years, with more than 85% of participants being assessed between the ages of 17 and 18. Although 84% of the adolescents were born in Israel, substantial heterogeneity in their origins was observed, reflecting the diversity of the worldwide Jewish immigration to Israel. Most participants (80%) had a full high school education.

The follow-up characteristics across the BMI categories are shown in Table 2. The median follow-up since 1981 (18.4 years) was lower in those with a higher BMI, a consequence of the later onset of the increasing prevalence of overweight and obesity in our cohort in the 1980s (15). During 42,297,007 person-years of follow-up, there were 226 deaths attributed to kidney injury out of a total of 32,127 deaths that were recorded in the cohort between 1981 and 2011;

TABLE 1 Baseline characteristics of the study participants by BMI categories

	BMI categories (kg/m ²)					Total
	<18.5	18.5-21.99	22.0-24.99	25.0-29.99	≥30.0	
<i>N</i> participants	315,066	1,126,717	549,602	239,423	63,331	2,294,139
% Males	59	60	59	58	60	60
Age ± SD (y)	17.3 ± 0.4	17.3 ± 0.4	17.3 ± 0.4	17.3 ± 0.4	17.4 ± 0.5	17.3 ± 0.4
Height ± SD (boys) (cm)	173.2 ± 7.0	173.4 ± 6.8	173.7 ± 6.8	174.0 ± 6.9	174.0 ± 7.5	173.5 ± 6.8
Height ± SD (girls) (cm)	162.8 ± 6.3	162.1 ± 6.0	161.8 ± 6.1	161.8 ± 6.2	162.3 ± 6.4	162.1 ± 6.1
Mean BMI ± SD (kg/m ²)	17.5 ± 0.8	20.3 ± 1.0	23.3 ± 0.8	26.8 ± 1.4	32.9 ± 2.7	21.6 ± 3.4
12 years schooling (%)	79	80	81	81	81	80
Low residential SES	24	25	23	25	27	24
Born in Israel	85	84	84	84	85	84
Country of origin (%)						
Israel	6	6	6	6	6	6
USSR	13	12	13	14	14	13
Asia	24	24	25	24	25	24
Africa	23	24	24	24	25	24
Europe	30	32	31	31	29	31
Ethiopia	4	1	1	1	1	1

Countries of origin classified by the father's or grandfather's country of birth (if the father was born in Israel) and grouped into the following six geographical areas: former Union of Soviet Socialist Republics (USSR), Asia (not including Asian USSR), Africa (predominantly North Africa, excluding South Africa and Ethiopia), Europe (and the following countries of emigration from Europe: North and South America, South Africa, Australia, and New Zealand, and excluding European USSR), Ethiopia, and Israel. SES, socioeconomic status.

TABLE 2 Association between BMI and mortality attributed to kidney disease

	BMI (kg/m ²)					
	<18.5	18.50-21.99	22.0-24.99	25.0-29.99	≥30.0	Total ^a
<i>Kidney disease-related mortality: all participants (226 deaths)</i>						
<i>N</i> participants	19	96	55	37	19	226
Deaths in men	17	81	45	29	15	187
Follow-up (y) (mean ± SD)	18.2 ± 9.6	19.1 ± 9.8	18.5 ± 9.9	16.7 ± 9.8	13.6 ± 9.1	18.4 ± 9.8
Median follow-up (y) (25th, 75th)	18.1 (10.0, 28.1)	19.4 (10.6, 30.1)	18.6 (9.8, 29.5)	16.0 (8.1, 25.7)	11.7 (6.0, 19.8)	18.4 (9.9, 29.10)
Cumulative follow-up (person-years)	5,747,457	21,507,635	10,177,773	4,004,835	859,307	42,297,007
Crude incidence (death/10 ⁵ person-years)	0.331	0.446	0.540	0.923	2.211	0.534
HR (unadjusted) ^b	0.80	1	1.24	2.36	7.30	1.15
95% CI	0.49-1.30	(Ref)	0.89-1.72	1.61-3.44	4.45-11.95	1.11-1.19
<i>P</i>	0.366		0.210	9.52 × 10 ^{−6}	2.95 × 10 ^{−15}	4.92 × 10 ^{−17}
HR (multivariable) ^b	0.84	1	1.32	2.64	8.40	1.16
95% CI	0.51-1.37	(Ref)	0.95-1.84	1.80-3.87	5.11-13.81	1.13-1.20
<i>P</i>	0.486		0.101	6.62 × 10 ^{−7}	5.04 × 10 ^{−17}	2.03 × 10 ^{−19}
<i>Kidney disease-related mortality among participants with unimpaired health (159 deaths)</i>						
<i>N</i> deaths	12	71	43	25	8	159
Deaths in men	10	61	34	19	7	131
HR (multivariable)	0.87	1	1.40	2.62	7.30	1.16
95% CI	0.47-1.61	(Ref)	0.96-2.05	1.66-4.15	3.50-15.23	1.11-1.21
<i>P</i>	0.657		0.080	3.93 × 10 ^{−5}	1.17 × 10 ^{−7}	5.51 × 10 ^{−10}

Person-time incidence rate per 100,000 person-years of follow-up shown. Association assessed with Cox models adjusted for age, sex, birth year, education, and country of origin. For all outcomes, there was a significant trend across BMI categories ($P \leq 0.001$).

^aValues reflect total sum of BMI categories or HR ($\pm 95\%$ CI) for death attributed to kidney disease for every unit increment in BMI.

^bNo interaction between sex and BMI for kidney disease-related mortality in unadjusted or multivariable model ($P = 0.646$ and 0.712 , respectively).

HR, hazard ratio.

83% of these ($n = 187$) occurred in men; 5.8%, 18.6%, 49.6%, and 97.6% occurred before age 30, 40, 50, and 60 years, respectively, with 50.4% of these deaths ($n = 114$) occurring from 2006 onward. The mortality rate increased steeply across age groups (Supporting Information Table S1). The mean age at death attributed to kidney disease was 48.1 ± 9.2 , reflecting the relatively young ages of the cohort at end of the follow-up.

Table 2 delineates follow-up time, crude person-year incidence rates, and adjusted Cox analysis across BMI groups. There was a graded increase in incidence, generally commencing from the low-normal BMI category ($18.5 \leq \text{BMI} < 22.0$) onward. When analyzed as a continuous variable, BMI was associated with an unadjusted HR of 1.15 (1.11-1.19; $P = 4.9 \times 10^{-17}$) per unit increment in BMI for kidney disease mortality that persisted in the multivariable model: 1.16 (1.13-1.20; $P = 2.0 \times 10^{-19}$). When BMI was treated as categorical variable, the following high adjusted HRs for kidney disease mortality were evident for overweight and, particularly, adolescents with obesity: 2.6 (1.8-3.9) and 8.4 (5.1-13.8), respectively (with the low-normal BMI as the reference). BMI of 22.0 to 24.99 was associated with borderline significance (HR = 1.3; 95% CI: 1.0-1.8; $P = 0.1$). Figure 2 shows the Cox survival function of the study outcomes. Multivariable spline models indicated a minimum risk for renal mortality at BMI of 18.6, with a significantly increased risk seen above adolescent BMI values of 22.8 (Supporting Information Figure S1).

A series of sensitivity analyses was conducted. The restriction of the study sample to those with unimpaired health (Table 2) had little effect on the risk estimates among adolescents with overweight and obesity, thereby minimizing the possibility that concurrent medical conditions biased the association between BMI and kidney mortality. Risk estimates among adolescents with overweight and obesity persisted when the cohort was stratified by the age of death, but they were slightly stronger among those who died after the age of 50 years than those who died before (overweight, HR = 3.3 vs. 2.1; obesity, HR = 10.5 vs. 6.7) (Supporting Information Table S2). Similarly, results generally persisted when the cohort was stratified by the period of death, although risk estimates were slightly higher among adolescents with BMI in the high-normal, overweight, and obesity range for deaths occurring from 2006 onward than for the earlier period (HRs of 1.5, 3.0, and 8.8 vs. 1.1, 2.2, and 7.9, respectively) (Supporting Information Table S2). Limiting the analysis to participants enrolled in the study from the time when blood pressure data were available (1977 onward) yielded an unchanged risk when mean blood pressure was added to the multivariable model (52 deaths; HR = 1.16 per unit increment in BMI; 95% CI: 1.13-1.20). The association between BMI and kidney disease mortality also persisted when the latter was stratified by acute, chronic, or unspecified kidney injury. The computed HRs were 1.13 (1.05-1.21; $P = 0.001$), 1.19 (1.14-1.24; $P = 4.9 \times 10^{-15}$), and 1.15 (1.06-1.24; $P = 3.8 \times 10^{-5}$) per unit increment in BMI, respectively, and 2.1 (1.0-4.5;

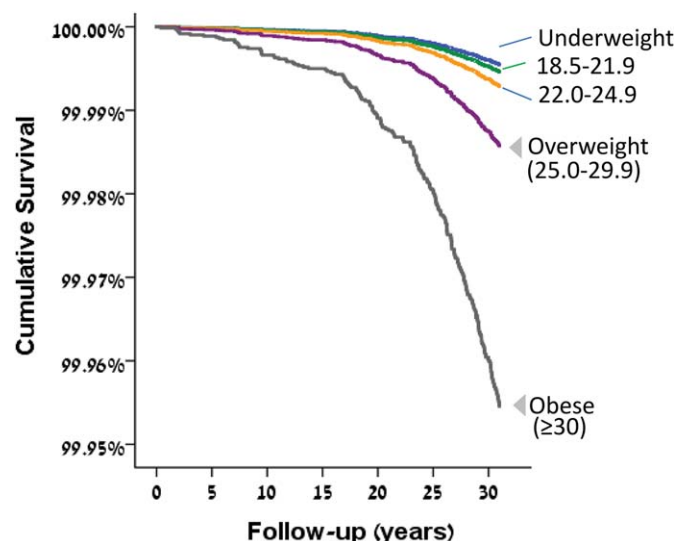


Figure 2 Relationship between BMI at adolescence and kidney disease survival. Multivariable Cox survival functions by BMI groupings are shown for mortality attributed to kidney disease. Models were adjusted for sex, age, birth year, education, and country of origin. [Color figure can be viewed at wileyonlinelibrary.com]

$P = 0.048$), 3.1 (1.8-5.2; $P = 2.1 \times 10^{-6}$), and 2.3 (1.0-5.5; $P = 0.052$) among adolescents with overweight compared with the low-normal BMI group, respectively.

Discussion

In this large nationwide cohort, we report on the association between late adolescent BMI and adult kidney disease-related mortality. The main findings of this study suggest that overweight and obesity increase the future risk for kidney disease-related mortality in apparently healthy teenagers with HRs of 2.7 and 8.4, respectively. Specifically, we observed a 16% increased risk for every unit increment in BMI. These findings persisted in a series of sensitivity analyses including stratification of kidney mortality because of acute or chronic kidney disease. Although the HRs for overweight- and obesity-related kidney mortality were high, the actual incidence of kidney-related mortality in our relatively young cohort was low but will increase sharply as the cohort ages. Interestingly, these point estimates were higher than those reported in this cohort for total cardiovascular (overweight, 2.3 [2.0-2.6]; obesity, 3.5 [2.9-4.1]; reference, BMI percentiles 5th-25th) (14) and cardiovascular cause-specific mortality (between 9% and 15.7% increased risk for the specific outcomes for every unit increment in BMI) (18).

In line with our findings, data from the Global Burden of Diseases project showed that approximately one-fourth of kidney mortality can be attributed to adulthood overweight and obesity (19). Also, a report that linked adult BMI with cause-specific mortality in a collaborative analyses of 57 studies showed a HR of 1.6 (95% CI: 1.3-2.0) for death because of renal disease per 5 BMI units in the range of > 25 (20). This value is modestly lower than the HR equivalent in our study (1.9; 95% CI: 1.7-2.0). The higher risk estimate reported here may reflect the stronger association between obesity and mortality and CKD incidence (11) at younger ages.

An earlier investigation from this cohort (12), as well as others (21), established the association between adolescent overweight and obesity and the risk of end-stage renal disease, a less frequent outcome than mortality among CKD patients (22,23), especially among young adults (24). For this cohort, the adjusted risk estimates among adolescents with overweight and obesity for incident end-stage renal disease were 3.0 and 6.9, respectively, similar to the values reported here. Thus, having obesity may not only lead to CKD, but also probably an increased risk of CKD-related death once established CKD exists. This contradicts the notion of a protective role of obesity suggested by previous studies that highlighted the “obesity paradox in CKD patients” (5).

Adolescent BMI is clearly a risk marker, but is it a causal risk factor? We were unable to control for adolescent risk factors, such as smoking, exercise, and physical fitness (25), or for adult BMI. We consider two pathways by which adolescent BMI might influence kidney outcomes. First, adolescent BMI has been causally associated with multiple cardiometabolic risk markers, as demonstrated by Mendelian randomization studies, even within the nonobese BMI range (26). This is consistent with our observation that the risk estimate for the association between BMI and kidney-related mortality persisted when the study sample was limited to those with unimpaired health. Young prepubertal children with obesity already present significantly lower glomerular filtration rate estimates, suggesting some degree of adiposity-induced renal impairment (27). Adolescents with severe obesity were shown to have increased urinary excretion of kidney injury biomarkers that may persist years after significant weight loss induced by bariatric surgery (28), with higher rates of renal complications following elective procedures (29). Furthermore, early exposure to obesity at age 20 was associated with a higher risk for future CKD (11) or cause-specific mortality (30) than that conferred by the onset of obesity later in life. This may relate to a longer duration of exposure to obesity from adolescence than observed in adulthood. Second, as BMI shows tracking across age (31), and strongly from age 18 (32), it is likely that adulthood BMI mediates the association of adolescent BMI with kidney disease, at least in part. This pathway might explain the borderline association of high-normal BMI with kidney-related mortality. As adult measures of BMI were unavailable to us, we were unable to address this question.

Our study has certain limitations. First, as noted, we could not assess an independent effect of adolescent BMI in the absence of adult measures of BMI. Second, data on cause-specific death were unavailable to us for the first 14 years of this cohort. Yet deaths attributed to kidney disease before age 30 accounted for only 5.8% of the total kidney-related deaths recorded during the 31-year follow-up period, indicating that the potential effect of these missing data should be minimal. Third, we were unable to account for cardiovascular lifestyle risk factors in adolescence, such as physical activity and smoking, that may confound the BMI association. However, adjustment for sociodemographic variables that were previously shown to be strongly related to smoking in a subpopulation of this cohort (33) did not attenuate the association. Fourth, the mean age of kidney-related deaths was 48.1 years as our follow-up encompassed young and midadulthood, thereby reflecting mortality in young adulthood. We show, nevertheless, that cause-specific mortality in our cohort is representative of the Israeli population (13,14) and is also similar to that reported by other cohorts with similar ages at enrollment and duration of follow-up (34). Fifth, waist circumference, which reflects abdominal adiposity and has been shown to be associated with mortality (35) and CKD incidence (36)

independently of BMI, was not measured in our cohort. Nevertheless, BMI is a well-studied measure that is currently recommended by the US Preventive Services Task Force as the screening measure of choice for childhood and adolescent obesity (37). Finally, the outcome of the study, kidney disease-related mortality, may be underestimated by deaths attributed to cardiovascular or diabetic causes (38). Nevertheless, similar to the United States and other high-income countries, kidney-related mortality is ranked among the 10 leading causes of death in the general population (39,40). The strengths of the study include the standardized measurement rather than recalled height and weight (41), the large sample size that provides stable estimates, and the extended follow-up.

In conclusion, in this large nationwide cohort, we confirmed the association between adolescent overweight and obesity and the deleterious consequences related to kidney disease. Our data suggest that BMI among adolescents with unimpaired health is a strong predictor of kidney mortality because of both acute and chronic kidney injury in early and midadulthood, with increased risk already seen within the currently accepted normal BMI range. The secular increase in overweight and obesity prevalence may account for a substantial and increasing burden of kidney-related mortality. **O**

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References

- Ogden CL, Carroll MD, Lawman HG, et al. Trends in obesity prevalence among children and adolescents in the United States, 1988-1994 through 2013-2014. *JAMA* 2016;315:2292-2299.
- GBD 2015 Obesity Collaborators; Afshin A, Forouzanfar MH, Reitsma MB, et al. Health effects of overweight and obesity in 195 countries over 25 years. *New Engl J Med* 2017;377:13-27.
- GBD compare data visualization. Institute for Health Metrics and Evaluation website. <https://vizhub.healthdata.org/>. Accessed January 1, 2015.
- Stengel B, Tarver-Carr ME, Powe NR, Eberhardt MS, Brancati FL. Lifestyle factors, obesity and the risk of chronic kidney disease. *Epidemiology* 2003;14:479-487.
- Kalantar-Zadeh K, Rhee CM, Chou J, et al. The obesity paradox in kidney disease: how to reconcile it with obesity management. *Kidney Int Rep* 2017;2:271-281.
- Iseki K, Ikemiya Y, Kinjo K, Inoue T, Iseki C, Takishita S. Body mass index and the risk of development of end-stage renal disease in a screened cohort. *Kidney Int* 2004;65:1870-1876.
- Reynolds K, Gu D, Muntner P, et al. Body mass index and risk of ESRD in China. *Am J Kid Dis* 2007;50:754-764.
- Fox CS, Larson MG, Leip EP, Culleton B, Wilson PW, Levy D. Predictors of new-onset kidney disease in a community-based population. *JAMA* 2004;291:844-850.
- van Dam RM, Willett WC, Manson JE, Hu FB. The relationship between overweight in adolescence and premature death in women. *Ann Intern Med* 2006;145:91-97.
- Tirosh A, Shai I, Afek A, et al. Adolescent BMI trajectory and risk of diabetes versus coronary disease. *New Engl J Med* 2011;364:1315-1325.
- Silverwood RJ, Pierce M, Thomas C, et al. Association between younger age when first overweight and increased risk for CKD. *J Am Soc Nephrol* 2013;24:813-821.
- Vivante A, Golan E, Tzur D, et al. Body mass index in 1.2 million adolescents and risk for end-stage renal disease. *Arch Intern Med* 2012;172:1644-1650.
- Twig G, Tirosh A, Leiba A, et al. BMI at age 17 years and diabetes mortality in midlife: a nationwide cohort of 2.3 million adolescents. *Diabetes Care* 2016;39:1996-2003.
- Twig G, Yaniv G, Levine H, et al. Body-mass index in 2.3 million adolescents and cardiovascular death in adulthood. *New Engl J Med* 2016;374:2430-2440.
- Twig G, Afek A, Shamiss A, et al. Adolescence BMI and trends in adulthood mortality: a study of 2.16 million adolescents. *J Clin Endocrinol Metab* 2014;99:2095-2103.
- Leiba A, Twig G, Vivante A, et al. Prehypertension among 2.19 million adolescents and future risk for end-stage renal disease. *J Hypertens* 2017;35:1290-1296.
- Greulich WW, Pyle SI. *Radiographic Atlas of Skeletal Development of the Hand and Wrist*. Palo Alto, CA: Stanford University Press; 1959.
- Twig G, Ben-Ami Shor D, Furer A, et al. Adolescent body mass index and cardiovascular disease-specific mortality by midlife. *J Clin Endocrinol Metab* 2017;102:3011-3020.
- Global Burden of Metabolic Risk Factors for Chronic Diseases Collaboration. Cardiovascular disease, chronic kidney disease, and diabetes mortality burden of cardiometabolic risk factors from 1980 to 2010: a comparative risk assessment. *Lancet Diabetes Endocrinol* 2014;2:634-647.
- Whitlock G, Lewington S, Sherliker P, et al. Body-mass index and cause-specific mortality in 900 000 adults: collaborative analyses of 57 prospective studies. *Lancet* 2009;373:1083-1096.
- Silverwood RJ, Pierce M, Hardy R, et al. Early-life overweight trajectory and CKD in the 1946 British birth cohort study. *Am J Kidney Dis* 2013;62:276-284.
- Drey N, Roderick P, Mullee M, Rogerson M. A population-based study of the incidence and outcomes of diagnosed chronic kidney disease. *Am J Kidney Dis* 2003;42:677-684.
- Keith DS, Nichols GA, Gullion CM, Brown JB, Smith DH. Longitudinal follow-up and outcomes among a population with chronic kidney disease in a large managed care organization. *Arch Intern Med* 2004;164:659-663.
- Chronic Kidney Disease Prognosis Consortium; Matsushita K, van der Velde M, Astor BC, et al. Association of estimated glomerular filtration rate and albuminuria with all-cause and cardiovascular mortality in general population cohorts: a collaborative meta-analysis. *Lancet* 2010;375:2073-2081.
- Shah RV, Murthy VL, Colangelo LA, et al. Association of fitness in young adulthood with survival and cardiovascular risk: The Coronary Artery Risk Development in Young Adults (CARDIA) Study. *JAMA Intern Med* 2016;176:87-95.
- Wurtz P, Wang Q, Kangas AJ, et al. Metabolic signatures of adiposity in young adults: Mendelian randomization analysis and effects of weight change. *PLoS Med* 2014;11:e1001765. doi:10.1371/journal.pmed.1001765
- Correia-Costa L, Afonso AC, Schaefer F, et al. Decreased renal function in overweight and obese prepubertal children. *Pediatr Res* 2015;78:436-444.
- Xiao N, Devarajan P, Inge TH, Jenkins TM, Bennett M, Mitsnefes MM. Subclinical kidney injury before and 1 year after bariatric surgery among adolescents with severe obesity. *Obesity (Silver Spring)* 2015;23:1234-1238.
- Calder CL, Ortega G, Vij A, et al. Morbid obesity is an independent risk factor for postoperative renal dysfunction in young adults: a review of the American College of Surgeons National Surgical Quality Improvement Program database. *Am J Surg* 2016;211:772-777.
- Abdullah A, Wolfe R, Stoelwinder JU, et al. The number of years lived with obesity and the risk of all-cause and cause-specific mortality. *Int J Epidemiol* 2011;40:985-996.
- Jeffreys M, McCarron P, Gunnell D, McEwen J, Smith GD. Body mass index in early and mid-adulthood, and subsequent mortality: a historical cohort study. *Int J Obes Relat Metab Disord* 2003;27:1391-1397.
- Kvaavik E, Tell GS, Klepp KI. Predictors and tracking of body mass index from adolescence into adulthood: follow-up of 18 to 20 years in the Oslo Youth Study. *Arch Pediatr Adolesc Med* 2003;157:1212-1218.
- Kark JD, Laor A. Cigarette smoking and educational level among young Israelis upon release from military service in 1988--a public health challenge. *Isr J Med Sci* 1992;28:33-37.
- Hogstrom G, Nordstrom A, Nordstrom P. Aerobic fitness in late adolescence and the risk of early death: a prospective cohort study of 1.3 million Swedish men. *Int J Epidemiol* 2016;45:1159-1168.
- Pischoon T, Boeing H, Hoffmann K, et al. General and abdominal adiposity and risk of death in Europe. *New Engl J Med* 2008;359:2105-2120.
- Odagiri K, Mizuta I, Yamamoto M, Miyazaki Y, Watanabe H, Uehara A. Waist to height ratio is an independent predictor for the incidence of chronic kidney disease. *PLoS One* 2014;9:e88873. doi:10.1371/journal.pone.0088873
- US Preventive Task Force; Grossman DC, Bibbins-Domingo K, Curry SJ, et al. Screening for obesity in children and adolescents: US Preventive Services Task Force recommendation statement. *JAMA* 2017;317:2417-2426.
- Rao C, Adair T, Bain C, Doi SA. Mortality from diabetic renal disease: a hidden epidemic. *Eur J Public Health* 2012;22:280-284.
- Kochanek KD, Murphy SL, Xu J, Tejada-Vera B. Deaths: final data for 2014. National Vital Statistics Reports, vol 65, no. 4. Hyattsville, MD: National Center for Health Statistics; 2016.
- Goldberger N, Aburbeh M, Haklai Z. Leading Causes of Death in Israel, 2001-2014. Jerusalem: State of Israel Ministry of Health; 2016.
- Keith SW, Fontaine KR, Pajewski NM, Mehta T, Allison DB. Use of self-reported height and weight biases the body mass index-mortality association. *Int J Obes (Lond)* 2011;35:401-408.